SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the different types of Shallow foundations used in practice.

Q.24 A R.C.C. beam 250 mm x 500 mm (overall) is reinforced with 4-25 mm. ∅ Effective cover is 40mm and effective span is 6.5m. FE 500 steel and M20 concrete is used in the beam section. Find the safe U.D.I. per meter which the beam can carry including and excluding self weight of the beam. Assume any other missing data.

Q.25 Design the reinforcement details of a simply supported slab resting on masonry walls to the following requirements.

a) Clear span = 3.00 m

b) Live load = 2.5 KN/m^2

Use M20 concrete and Fe-415 steel. Assume any other missing data.

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Roll No. ....

6th Sem/ Architectural, Architectural
(For Speech & Hearing Impaired)
Subject : Structure Systems-II

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Concrete is strong in

a) Compression b) Tension

c) Shear d) Torsion

Q.2 A foundation is called a shallow foundation if

a) The ratio of it's depth and breadth is less than 1

b) The ratio of its depth and breadth is more than 1

c) It's depth is less than 1m below the ground surface

d) None of these

Q.3 Design is based on

a) 7 days compressive strength

b) 14 days compressive strength

c) 28 days compressive strength

d) None of the above

Q.4 When the slenderness ratio of column exceeds 12 then it is termed as

a) Long b) Short

c) Medium d) None of these

Q.5 Splicing of bars at one section should not exceed

a) 30% b) 40%

c) 50% d) 60%

Q.6 Two ways slabs are provided if the Long span / Short span ratio of

a) Less than2 b) Greater than2

c) Equal to 2 d) None of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Weight of steel is taken as _______.

Q.8 The type of foundation most suitable for bridge is _____.

Q.9 The maximum strain in concrete is assumed as _____ in bending.

Q.10 Buckling in long column is ______.

Q.11 In cantilever beams, the main reinforcement is provided _____ the neutral axis.

Q.12 The structural member which have very less thickness as compared to other dimensions is called as _______.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Why steel is suitable as best reinforcing materials in concrete?

Q.14 What are the disadvantages of R.C.C. (any four)

Q.15 Write short note on Deep foundation.

Q.16 Differentiate between long and short Column. (any four)

Q.17 What are the functions of longitudinal reinforcements in Column?

Q.18 What are the Purposes of designing Columns?

Q.19 What are the functions of Beam in a structure?

Q.20 State the conditions under which doubly reinforced beams are provided.

Q.21 Why distribution steel is provided in one way slabs?

Q.22 Why two way slabs are considered economical?